JIGAWA STATE INTER-SCHOOL SPELLING BEE CHAMPIONSHIP

OFFICIAL SYSTEM ARCHITECTURE & TOURNAMENT OPERATIONS BRIEFING

SPONSOR	ORGANIZER	DEVELOPER	PROJECT LEAD
Jigawa State Government Ministry of Education	J-YEDA Youth Educational Assoc.	KowaguruTech Company Lead Engineering Team	Aliyu Yahaya Masset Competition Operations

1. Executive Summary & Platform Overview

This official briefing details the enterprise web application engineered by **KowaguruTech** for the **Jigawa State Inter-School Spelling Bee Championship**, sponsored by the **Jigawa State Ministry of Education** and organized by **J-YEDA**. Built with Node.js, Socket.IO WebSockets, and a robust PostgreSQL 18 database, the ecosystem eliminates paper scorekeeping with synchronized real-time screens across the auditorium. The system operates 100% offline across hall Wi-Fi, providing instant scoring, automated qualification checks, and strict audit integrity.

2. Official Tournament Stages & Mathematical Progression

In accordance with official directives from Project Lead **Aliyu Yahaya Masset**, the tournament is executed across **four qualification stages** and a **reserve pool**. The system strictly monitors speller attempts and enforces advancement thresholds:

Stage	Word Range	Words / Speller	Total Word Math	Advancement / Elimination Rule
Round 1 **(First Round)**	Q1 – Q100 *Simplest Tier*	4 Words	18 × 4 = **72 words**	Must score ≥ **3 correct out of 4** to advance to Round 2. Contestants with < 3 correct are eliminated from progression.
Round 2 **(Second Round)**	Q101 – Q200 *Intermediate*	3 Words	18 × 3 = **54 words**	Must score ≥ **2 correct out of 3** to advance to the Semi-Final. Contestants with < 2 correct are eliminated.
Round 3 **(Semi-Final)**	Q201 – Q300 *Advanced*	3 Words	18 × 3 = **54 words**	**100% Accuracy Required:** Contestants must answer **ALL 3 words correct** to qualify for the Championship Final.
Round 4 **(Final Round)**	Q301 – Q400 *Championship*	Sudden Death (2 Words Base)	18 × 2 = **36 words**	**Sudden Death Knockout:** 'Spell & stay • Misspell & go.' The last remaining uneliminated contestant is crowned Champion.
Reserve Pool **(Tie-Breakers)**	Q401 – Q500 *Master Tier*	1 Word per trial	**100 Reserve words**	If all finalists fail in a cycle, single-word trials continue from the 100 reserve words until the champion is ascertained.

3. Official 500-Word Bank Architecture

The official J-YEDA competition list (**500 words**) extracted from '*Jigawa Spelling Bee JSS 500 Words 4 Rounds.pdf*' has been permanently loaded into PostgreSQL. It includes phonetic guides, parts of speech, and curriculum subjects (English, Mathematics, Basic Science, Agriculture, Digital Technology, Business Studies, Social Studies, Arts, etc.). **Per user specification, all screens display only the active round's questions (100 per round) rather than an overwhelming 500-word wall.**

4. Participating Schools & Contestants Roster (18 Slots)

The competition includes **6 schools with 3 contestants each (18 total spellers)**. The 14 confirmed names are live in the database; 4 structured placeholders are primed for instant one-click updating as soon as field organizers transmit the remaining details:

ID	Contestant Name	School Name	LGA / Zone	Category	Roster Status
#001	**Humaira Sani**	MBPS Gumel	Gumel	JSS	**Confirmed Active**
#002	**Khadija Lawan**	MBPS Gumel	Gumel	JSS	**Confirmed Active**
#003	**Adama Ya'u**	MBPS Gumel	Gumel	JSS	**Confirmed Active**
#004	**Musa Yakubu**	MBPS Bulangu	Bulangu	JSS	**Confirmed Active**
#005	**Ibrahim Yakubu**	MBPS Bulangu	Bulangu	JSS	**Confirmed Active**
#006	**Abdurrahid Muhammad**	MBPS Bulangu	Bulangu	JSS	**Confirmed Active**
#007	**Fatima Munkaila Alhassan**	MBPS Kudai Dutse	Dutse	JSS	**Confirmed Active**
#008	**Ummusalma Baffa**	MBPS Kudai Dutse	Dutse	JSS	**Confirmed Active**
#009	**Maimuna Muaazu**	MBPS Kudai Dutse	Dutse	JSS	**Confirmed Active**
#010	**Ummusalma Ya'u**	Nomadic Dutse	Dutse	JSS	**Confirmed Active**
#011	**Zuwairiyya Sulaiman Haruna**	Nomadic Dutse	Dutse	JSS	**Confirmed Active**
#012	*[Nomadic Speller 3]*	Nomadic Dutse	Dutse	JSS	Pending Name Update
#013	**Ahmad Usman**	Mega School Dutse	Dutse	JSS	**Confirmed Active**
#014	**Zainab Yahaya**	Mega School Dutse	Dutse	JSS	**Confirmed Active**
#015	**Sakina Abdulaziz**	Mega School Dutse	Dutse	JSS	**Confirmed Active**
#016	*[Dangan Tsaure Speller 1]*	JSS Dangan Tsaure	Dutse	JSS	Pending Name Update
#017	*[Dangan Tsaure Speller 2]*	JSS Dangan Tsaure	Dutse	JSS	Pending Name Update
#018	*[Dangan Tsaure Speller 3]*	JSS Dangan Tsaure	Dutse	JSS	Pending Name Update

5. Synchronized Multi-Screen Environment

■■■■ Judges & Pronouncer Cockpit (/judge)
Private console with Web Speech audio pronunciation, round-locked question picker (100 questions per round), live attempt trackers (e.g. *'Attempt 1 of 4 • 3 needed'*), hotkeys (F5 pronounce, Space timer, Enter correct, Esc incorrect), and live roster tracking.

■■ Auditorium Stage Projector (/projector)
High-contrast stage view for spectators and dignitaries. Features official EGuru branding, active round banner, giant 60s countdown circle, real-time live letter streaming, and instant GREEN / RED adjudication reveal banners.

■■■ Speller Podium Monitor (/speller)
1. Question Picker Mode: Displays only the 100 question numbers for the active round. Speller taps their choice, which glows green and marks 'chosen' for them to announce to the judge.
2. Active Turn Mode: Displays full-screen timer, contestant info, and guidance (*'Spell your word clearly into the microphone'*).

■ Ministerial Leaderboard & School Rankings (/leaderboard)
Live score standings across contestants and schools. Includes printable attestation certificates and sign-off blocks.

6. Auditorium Network Deployment & Wi-Fi Architecture

The application is hosted on the primary operations laptop on port 3000. All auditorium laptops, tablets, and projector computers connect seamlessly over the venue's local Wi-Fi without needing an internet connection:

Screen / Role	Local URL (Host Machine)	Auditorium Wi-Fi URL (Tablets / Laptops)
Main Portal Hub	http://localhost:3000/	http://192.168.43.36:3000/
Judges & Pronouncer Cockpit	http://localhost:3000/judge	http://192.168.43.36:3000/judge
Stage Projector Screen	http://localhost:3000/projector	http://192.168.43.36:3000/projector
Speller Podium Monitor	http://localhost:3000/speller	http://192.168.43.36:3000/speller
Ministerial Leaderboard	http://localhost:3000/leaderboard	http://192.168.43.36:3000/leaderboard

7. Key Engineering Upgrades Delivered in this Version

• **Round-Isolated Question Grid:** Replaced the full 500-word grid with round-specific 100-question views. In Round 1, spellers see strictly Questions 1–100. In Round 2, Questions 101–200, and so forth.

• **Automated Progression Engine:** System enforces multi-attempt qualification rules (Round 1: 4 words, 3 to pass; Round 2: 3 words, 2 to pass; Semi-Final: 3 words, 3 to pass; Final: Sudden Death).

• **Official EGuru & KowaguruTech Palette:** Re-themed into Championship Red (`#cc1b1b`), Ebony Black (`#060808`), and Emerald Green (`#1a7a1a`) with the cartoon bee mascot logo.

• **PostgreSQL 18 Real-Time Sync:** All student scores, round history, and word statuses persist safely with zero data loss on unexpected power cuts or browser reloads.

8. Technical Endorsement & Attestation Sign-Off

Submitted by Systems Engineering Team:	Received & Confirmed by Project Lead:	Ministry of Education Attestation:
_____	_____	_____
KowaguruTech Company Lead Software Development Team	**Aliyu Yahaya Masset** Project Lead, J-YEDA Spelling Bee	**Official Representative** Jigawa State Ministry of Education